

Reducing Wait Times at Roots Natural Kitchen Using a Tandem Queueing Model

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Abstract—Long queues at fast food restaurants may lead to reduced customer satisfaction and increased pressure on employees during busy periods. This study develops a mathematical model for the in-store ordering process at Roots Natural Kitchen, where customers make their orders, prepare their bowls, and pay in a sequential manner. Through the observation of arrival times and service times during three lunchtime periods, this paper derives the parameters for the arrival and service rates, confirms the Markovian assumptions with Kolmogorov-Smirnov tests, and compares the analytic predictions of steady-state measures with a discrete-event simulation of the model. The combined model indicates that the arrival rate is 0.662 customers per minute, with order-taking being the bottleneck with a utilization of 0.778. Analytical Jackson Network Queue Model predicts the average system time of 7.49 minutes, agreeing well with the simulation result of 7.45 minutes and remaining close to the observed average of 6.39 minutes. The effect of hiring another employee to take orders is then analyzed, turning the first stage from M/M/1 to M/M/2. According to the Erlang-C formula, the proposed strategy would decrease the average system time to 3.60 minutes, representing a 52.0% decrease in time compared to the existing process, while a discrete-event simulation provides a more conservative estimate of 4.24 minutes.

Keywords—queueing theory, Jackson network, CTMC, discrete-event simulation, restaurant operations, Erlang-C

I. INTRODUCTION

Roots Natural Kitchen is a quick-service restaurant close to the University of Virginia. It suffers from visible congestion at lunch time. This is a sequential process: a customer enters the line, orders food, gets his bowl put together, pays, and exits. Since there is a division of labor among employees, the system does not comprise interchangeable parallel servers. Thus, it constitutes a tandem service network. This project was done as part of the Stochastic Decision Models course offered by Daniel Otero-Leon [1] at the University of Virginia. Specifically, our team had to collect data on a system, construct a stochastic model, test the assumptions, and propose an improvement. Therefore, we aim to describe and propose improvements to the process that cause congestion at Roots.

There are three contributions of this paper. First, we formulate the Roots process as a continuous-time Markov chain tandem network model and argue why a single M/M/3 model fails to properly capture its dynamics. Second, we estimate queueing parameters based on timestamps and test assumptions using distributions. Third, we compare the existing system against its improved variant where stage one employs two parallel workers.

II. SYSTEM AND DATA

A. Physical Process

The observed process consists of three stages. Stage one involves order taking. At stage one, the customer starts interacting with workers and communicates the requested bowl. Stage two is assembling the bowl. Stage three is payment. In other words, customers have to sequentially traverse all three stages in order. A worker at one stage cannot replace a worker at another stage.

The timestamps observed by the team consist of five measurements per arrival: entering the line, starting stage one, stage two, stage three, and leaving. Thus, the difference between leaving and entering is observed total time in the system. Time between arrivals and service-stage durations were calculated based on a cleaned timestamp sheet. The full repository of the project, including the spreadsheet with timestamps and the analysis Jupyter notebook, can be found here [2].

B. Data Collection Caveats

The data collection process took place in a crowded real-world environment where only limited number of people could observe and take notes. When multiple customers approached the counter at once, the batch of them was taken as a single arrival since it was difficult to track individual customers in a batch. The collected data also needed cleaning from timestamping mistakes. Therefore, while these changes make the data fit for analysis, they do not make them perfect records of all individual customers. More on limitations in Section VII.

The final dataset includes 147 arrival timestamps distributed between three observation windows (Days 1, 2, and 3): 50, 44, and 53, respectively. The combined sample thus yields sufficient observations to conduct initial queueing analysis and retains day-to-day variation.

III. QUEUEING MODEL

A. Tandem CTMC Formulation

Let the CTMC state be

$$X(t) = (n_1(t), n_2(t), n_3(t)), \quad (1)$$

where $n_i(t)$ is the number of customers at stage i , including any customer in service. External arrivals enter Stage 1 at rate λ . Service completion at Stage 1 moves a customer from Stage 1 to Stage 2 at rate μ_1 when $n_1 > 0$; completion at

Stage 2 moves a customer to Stage 3 at rate μ_2 when $n_2 > 0$; completion at Stage 3 removes a customer from the system at rate μ_3 when $n_3 > 0$.

This is a tandem network of three M/M/1 queues. The M/M/1 assumption is that external interarrival times are exponentially distributed and that each stage's service time is exponentially distributed. In an open network of M/M/1 queues with Poisson external arrivals, the steady-state distribution has product form when $\rho_i = \lambda/\mu_i < 1$ for all stages. Each stage can therefore be analyzed with standard M/M/1 formulas:

$$\rho_i = \frac{\lambda}{\mu_i}, \quad L_i = \frac{\rho_i}{1 - \rho_i}, \quad L_{q,i} = \frac{\rho_i^2}{1 - \rho_i}, \quad (2)$$

$$W_i = \frac{1}{\mu_i - \lambda}, \quad W_{q,i} = \frac{\rho_i}{\mu_i - \lambda}. \quad (3)$$

The total expected time in system is

$$W_{\text{sys}} = W_1 + W_2 + W_3. \quad (4)$$

B. Why Not M/M/3

The M/M/3 model assumes that there are three servers that are interchangeable, and all of them are served by one queue. This is not the case at Roots. A customer at Roots is unable to bypass the queueing process in taking an order, going straight into payment. Also, the server at Roots who processes payments cannot interchangeably take orders.

IV. PARAMETER ESTIMATION AND VALIDATION

A. Estimated Rates

For exponential interarrival and service times, the maximum-likelihood estimate of a rate is the reciprocal of the sample mean. Table I reports the estimated arrival rate, service rates, and utilizations by day and for the pooled sample.

TABLE I
ESTIMATED RATES AND UTILIZATIONS

Scenario	λ	μ_1	μ_2	μ_3	ρ_1	ρ_2	ρ_3
Day 1	0.585	0.833	2.142	1.130	0.702	0.273	0.517
Day 2	0.609	0.863	2.045	1.519	0.706	0.298	0.401
Day 3	0.825	0.860	2.279	1.360	0.958	0.362	0.606
Pooled	0.662	0.852	2.158	1.310	0.778	0.307	0.505

The pooled estimates identify Stage 1 as the bottleneck. Its utilization is 0.778, compared with 0.307 for assembly and 0.505 for payment. Day 3 is especially important because its estimated Stage 1 utilization is 0.958, which places the first stage close to the M/M/1 stability boundary.

B. Distributional Validation

To test whether the Markovian assumptions are reasonable, we fit exponential distributions to the interarrival samples and the three service-time samples. We then applied the Kolmogorov–Smirnov (KS) goodness-of-fit test. The results are shown in Table II and Fig. 1. All four samples fail to reject the exponential model at the 0.05 level. The coefficients of variation are also close to 1.0, which is the signature value for an exponential distribution.

TABLE II
POOLED EXPONENTIAL-FIT VALIDATION

Variable	n	Mean	Rate	CV	KS p
Interarrival	143	1.520	0.658	1.036	0.967
Stage 1 service	147	1.174	0.852	0.991	0.744
Stage 2 service	147	0.463	2.158	0.993	0.592
Stage 3 service	147	0.763	1.310	0.976	0.727

Distribution Fits: Empirical vs Exponential (Pooled, $n = 147$)

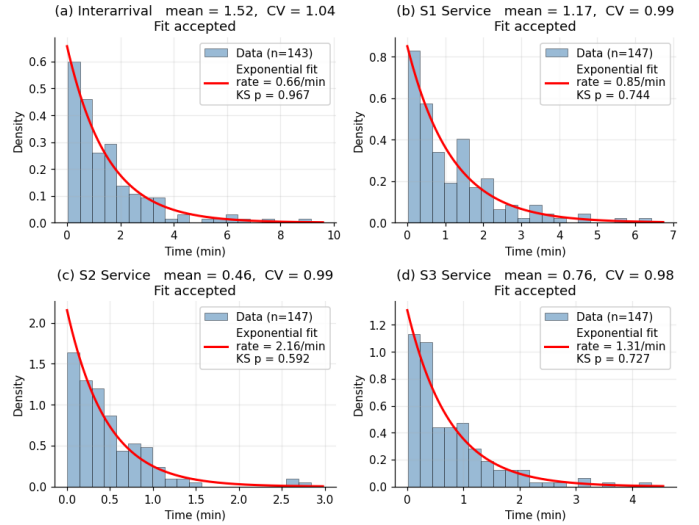


Fig. 1. Empirical interarrival and service-time distributions compared with fitted exponential distributions. All pooled KS tests fail to reject the exponential model at $\alpha = 0.05$.

C. Simulation Validation

Analytic formulas can hide implementation mistakes, so we validated the model using a discrete-event simulation (DES) of customer-by-customer movement through the three-stage tandem process. Each replication generates Poisson arrivals and exponential services using the estimated parameters, processes a large customer stream, discards an initial warmup period, and records mean time in system. Table III compares observed averages, DES output, and analytic predictions.

TABLE III
OBSERVED, SIMULATION, AND ANALYTIC MEAN SYSTEM TIME

Scenario	Observed W	DES W	Analytic W
Day 1	5.177 ± 0.422	6.473 ± 0.147	6.502
Day 2	4.802 ± 0.447	5.700 ± 0.152	5.731
Day 3	8.843 ± 0.621	29.931 ± 12.038	30.527
Pooled	6.387 ± 0.333	7.445 ± 0.234	7.490

The pooled analytic and DES estimates differ by less than 1%, which validates both the formula implementation and the simulation logic. Observed averages are lower than steady-state predictions because the collection windows are finite lunch intervals rather than long-run steady-state experiments. Day 3 also shows how sensitive M/M/1 waiting time becomes near saturation: as λ approaches μ_1 , the term $1/(\mu_1 - \lambda)$ grows rapidly.

V. BASE MODEL RESULTS

The base model confirms that the first stage dominates system performance. For the pooled model, Stage 1 contributes 5.278 minutes of the 7.490-minute analytic system time. Stage 2 contributes only 0.669 minutes, and Stage 3 contributes 1.543 minutes. In other words, improving assembly would have little effect unless order taking is also improved.

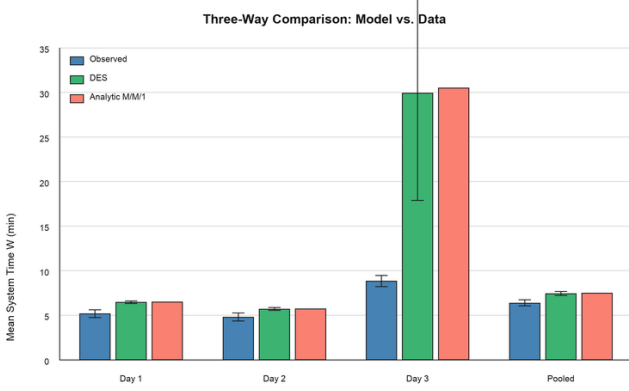


Fig. 2. Comparison of observed mean system time, DES output, and analytic M/M/1 tandem predictions. Day 3 illustrates near-saturation sensitivity at Stage 1.

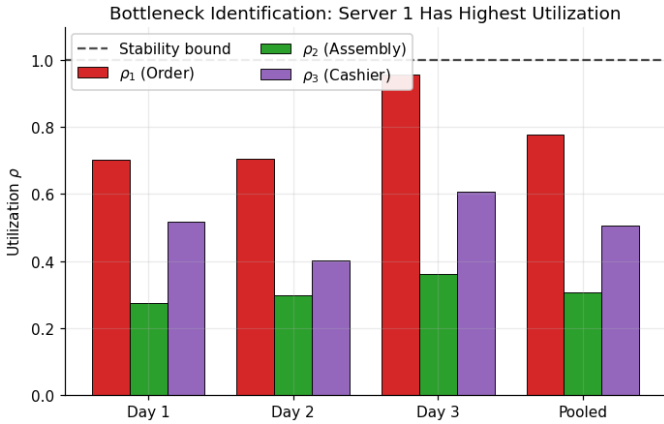


Fig. 3. Estimated utilization by stage. Stage 1 has the highest utilization in every scenario and is close to saturation on Day 3.

VI. IMPROVEMENT: PARALLEL ORDER TAKING

A. M/M/2 Stage 1

Since Stage 1 is the bottleneck, the proposed improvement is to add a second order-taker during peak demand. This converts Stage 1 from M/M/1 to M/M/2 while leaving assembly and payment as M/M/1 stages. For an M/M/ c queue, define $a = \lambda/\mu$ and $\rho = \lambda/(c\mu)$. The Erlang-C probability that an arrival must wait is

$$C(c, a) = \frac{\frac{a^c}{c!} \frac{1}{1-\rho}}{\sum_{k=0}^{c-1} \frac{a^k}{k!} + \frac{a^c}{c!} \frac{1}{1-\rho}}. \quad (5)$$

Then

$$L_q = C(c, a) \frac{\rho}{1-\rho}, \quad W_q = \frac{L_q}{\lambda}, \quad W = W_q + \frac{1}{\mu}. \quad (6)$$

B. Improvement Results

Table IV compares the base and improved designs. In the pooled case, the analytic system time decreases from 7.490 minutes to 3.595 minutes, a 52.0% reduction. Stage 1 alone falls from 5.278 minutes to 1.383 minutes. DES estimates are slightly more conservative but lead to the same decision: system time falls from 7.45 minutes to 4.24 minutes.

TABLE IV
EFFECT OF ADDING A SECOND ORDER-TAKER

Scenario	Base	Improved	DES Improved	Reduction
Day 1	6.502	3.844	4.47 ± 0.04	40.9%
Day 2	5.731	3.119	3.68 ± 0.03	45.6%
Day 3	30.527	4.065	4.86 ± 0.06	86.7%
Pooled	7.490	3.595	4.24 ± 0.04	52.0%

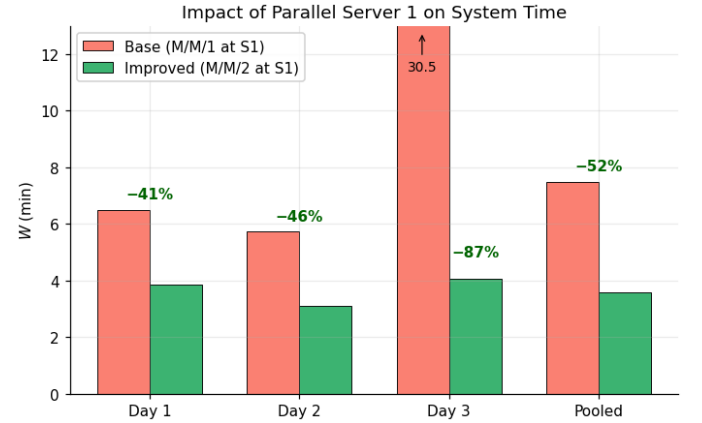


Fig. 4. Analytic mean system time before and after adding a second order-taker. The largest benefit appears on the near-saturated Day 3 scenario.

Sensitivity analysis further supports the intervention. As demand increases, the base M/M/1 Stage 1 approaches instability, causing total system time to grow sharply. The M/M/2 design remains much flatter over the tested demand range, as shown in Fig. 5. This means the second order-taker does more than reduce average wait today; it also makes the system more robust to peak loads.

VII. LIMITATIONS AND FUTURE WORK

The model captures the key sequence of the ordering process but streamlines various aspects of reality of the restaurant. Firstly, batch arrivals are treated as one unit. This approach helped to collect data, but it obscures variation within groups and might underestimate individual customer congestion. This point should be addressed in a future research by measuring the size of customer groups and developing a model of the queue with batch arrivals.

Secondly, the model assumes that arrival rates are constant. In reality, lunch service experiences a growth, peaks, and

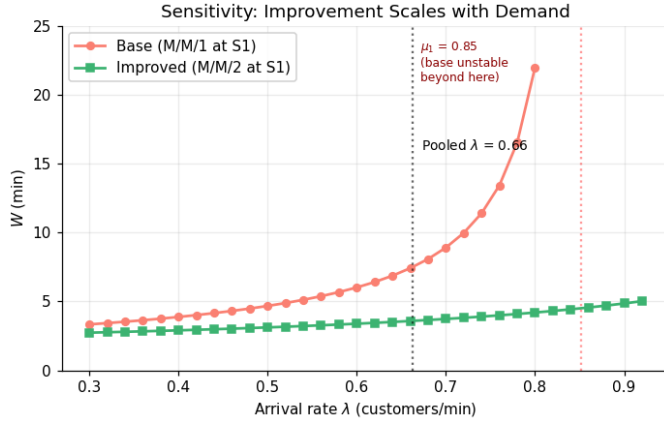


Fig. 5. Sensitivity of system time to arrival rate. The base design grows rapidly as demand approaches Stage 1 capacity, while the M/M/2 Stage 1 design remains stable over the tested range.

then falls. A varying arrival rate model like $M(t)/M/1$ should be considered to represent this situation better. Thirdly, the model uses infinite buffers between queues. However, blocking may occur in the actual service line when a downstream operator is not available for work, especially when the line is constrained by space limitations. Finally, we model each queue with an identical exponential service rate despite different order complexity, workers' experience, and customer payment behavior.

Finally, the proposed improvements account for time savings but not for the increase in labor costs. The next step should include a cost-effectiveness analysis to compare labor cost of hiring an additional order taker against savings in customer waiting time, number of served customers per hour, and potentially retained sales at peak hours.

VIII. CONCLUSION

This research has modeled Roots Natural Kitchen as a three-stage tandem queuing network based on data obtained through timestamps. We have used data on timestamps to estimate and validate our model and implement some improvements to it. According to the pooled model, order taking is a bottleneck because it operates with utilization 0.778. Assembly and payment processes are less utilized. Analytical results for the Jackson-network are confirmed by simulations, so we are confident about the implementation of the base model. Hiring an additional operator at the order-taking point Stage 1 is the most effective improvement: the time savings are 52.0%, and this will help avoid drastic increases in customer waiting time when approaching saturation.

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